

MEA Thermodynamics Research Notebook

MEA-Thermodynamics project

August 25, 2026

1 Neutral binary qualification with fixed CO₂–water induced association

1.1 Question and conditions

The practical neutral stage asks which source-backed MEA association family transfers between the two Cai MEA–water pressure levels under fixed Held 2B water, and whether the source CO₂–water interaction reproduces the independent Kiepe coexistence data. These calculations precede the nine-species reactive fit; no ionic or reaction parameter is estimated here.

1.2 Methods and sources

The MEA–water calculation fits one $k_{\text{MEA},\text{H}_2\text{O}}$ on the 101.33 kPa Cai series and evaluates the independent 66.66 kPa series. It compares the Baygi 2B, 3B, and 4C MEA families under the same Held water family. All 25 states use component log-fugacity closure and exact Engine derivatives checked against centered finite differences.

The CO₂–water calculation uses all 39 nonzero-composition rows from Kiepe et al. (2002), Table 1, DOI 10.1021/ie020154i. The molecular, association, and temperature-dependent binary parameters are from Pabsch et al. (2020), Tables 2 and 6, DOI 10.1021/acs.jced.0c00704. The calculation uses the declared liquid and incipient-vapor topology.

1.3 Results

Table 1: MEA–water family comparison. Log-RMSE is based on the natural-log fugacity-closure residual.

MEA family	$k_{\text{MEA},\text{H}_2\text{O}}$	Fit log-RMSE	Held log-RMSE	Held factor
2B	-0.058566	0.06883	0.15891	1.172
3B	-0.019567	0.05900	0.11123	1.118
4C	-0.036274	0.05244	0.12577	1.134

The 3B family gives the best pressure-level transfer and is retained. For CO₂–water, all 39 states evaluate. The log-pressure RMSE is 0.21228 and the median multiplicative error is 1.1703. The largest pressure and chemical-potential residuals are 2.36×10^{-9} and 1.85×10^{-8} , respectively.

1.4 Interpretation and limitations

The evidence supports freezing the Held-water/3B-MEA neutral family and the source CO₂–water temperature correlation before the electrostatic comparison. The MEA–water source reports composition resolution rather than a statistical covariance model, so the raw log closure

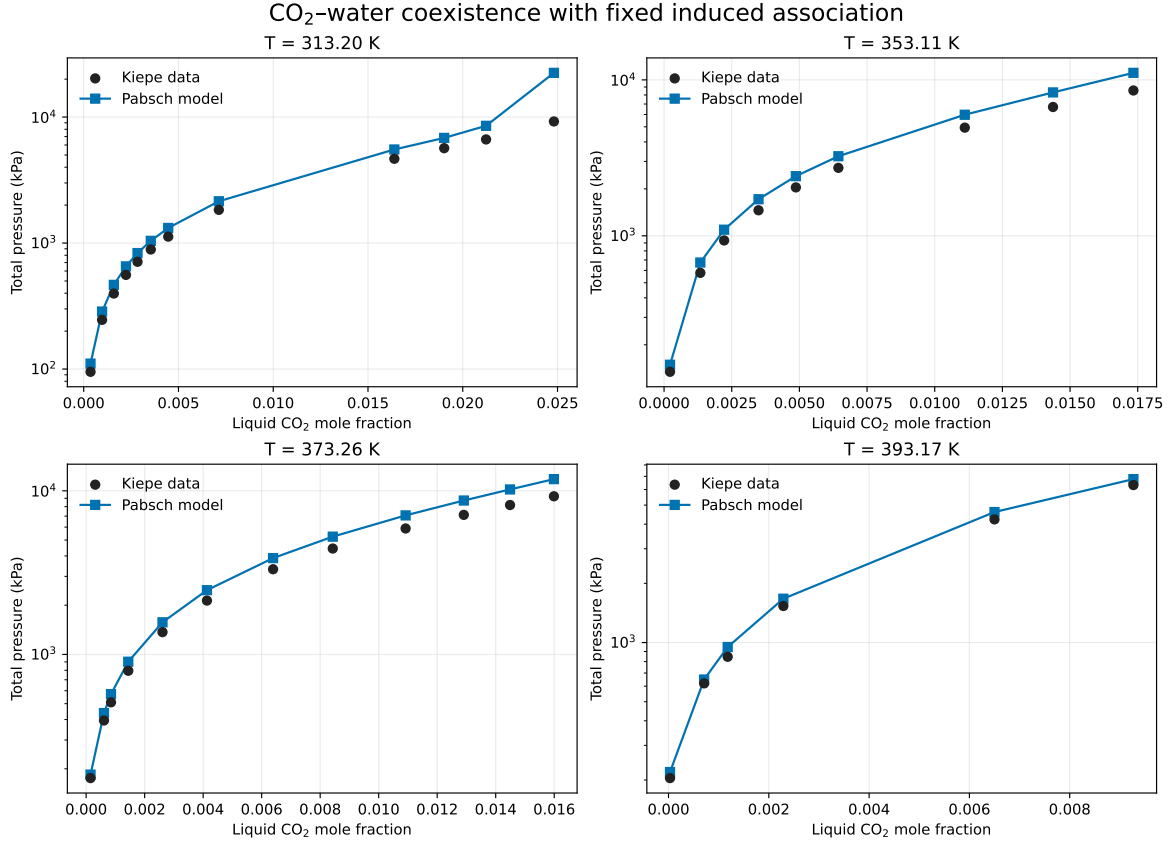


Figure 1: Kiepe observations and the source Pabsch CO₂–water calculation at four temperatures. Exact plotted rows are retained with the analysis.

and pressure-level transfer are more informative than the stringent resolution-normalized gate. The CO₂–water comparison reproduces the same data family used in the source parameterization; it establishes faithful implementation and practical agreement, not independent reserved validation. Neither binary calculation identifies ionic, reaction, or reactive-pressure parameters.

The exact inputs and outputs are linked from the pressure-first and CO₂–water analysis READMEs.

2 Born formulation and induced-association sensitivity

2.1 Question and conditions

The calculation crosses original Born versus shell-modified Born with ion-suppressed relative permittivity against CO₂–water induced association off/on. Temperature, pressure, unloaded MEA mass fraction, and CO₂ loadings are 313.15 K, 101 325 Pa, 30 wt%, and 0.11–0.99 mol CO₂/mol MEA. Neutral molecular parameters, five reaction correlations, and all other ionic parameters remain fixed.

2.2 Methods and sources

Original-Born cases use solvent-only relative permittivity and $c_{\text{shell}} = c_{\text{dielectric}} = 0$. Shell-modified-Born cases use $c_{\text{shell}} = c_{\text{dielectric}} = 1$ together with the fixed ion-fraction suppression coefficient 7.01. Six Jakobsen et al. (2005) speciation states are evaluated. The Amundsen et al. (2009) unloaded density of 1003.4 kg m^{-3} is used only as a liquid-density scale.

2.3 Fixed-parameter result

Table 2: Born-formulation by induced-association sensitivity.

Formulation	Evaluated	Density (kg/m ³)	Factor
Original Born, association off	2/6	1.087–1.125	2.531
Original Born, association on	2/6	1.087–1.125	2.531
Shell Born, ion-suppressed, association off	6/6	1065.5–1445.3	1.3235
Shell Born, ion-suppressed, association on	6/6	1065.5–1447.6	1.3238

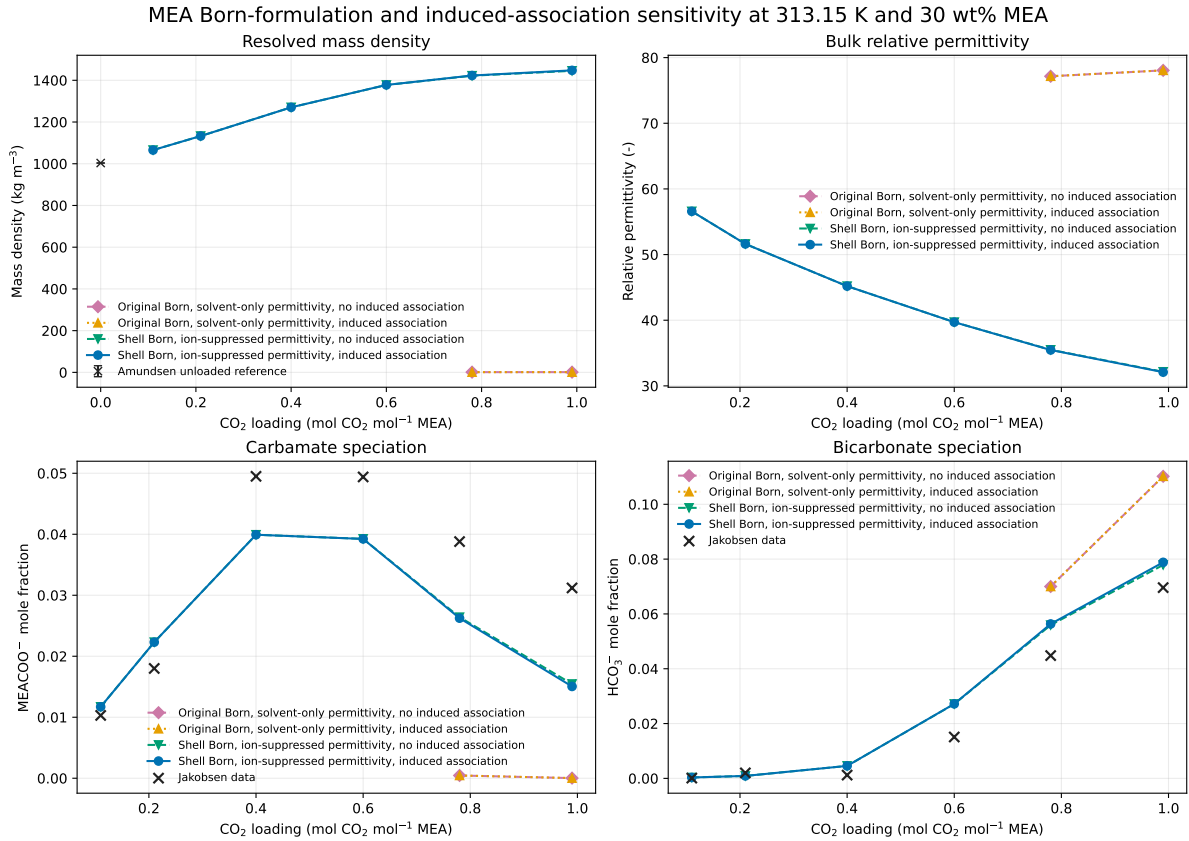


Figure 2: Resolved density, relative permittivity, and Jakobsen carbamate and bicarbonate comparisons. Exact plotted values are retained with the analysis.

Both original-Born cases fail four states and return vapor-density roots for the remaining two. Both shell-Born, ion-suppressed cases resolve all six liquid-density states. Induced association changes their density by at most 2.27 kg m^{-3} ; their fixed-parameter typical speciation factors are 1.3235 and 1.3238.

2.4 Two-ion-diameter result

Only the protonated-MEA and carbamate-anion segment diameters are fitted. The objective contains 32 unweighted natural-log residuals for direct-positive Jakobsen observations. Aggregate rows are excluded. Original-Born cases are not fitted because they supply no admissible liquid starting branch.

Both fits move the protonated-MEA diameter from $3.48509 \text{ \AA}$ to the exploratory lower bound and the carbamate-anion diameter from $3.53544 \text{ \AA}$ to approximately $3.07 \text{ \AA}$. The two-column

Table 3: Two-ion-diameter sensitivity fit under shell Born and ion-suppressed relative permittivity.

Association	σ_{MEA^+} (Å)	σ_{MEACOO^-} (Å)	log SSE	Factor
Off	2.00000	3.07165	22.811	1.477
On	2.00000	3.06644	23.303	1.442

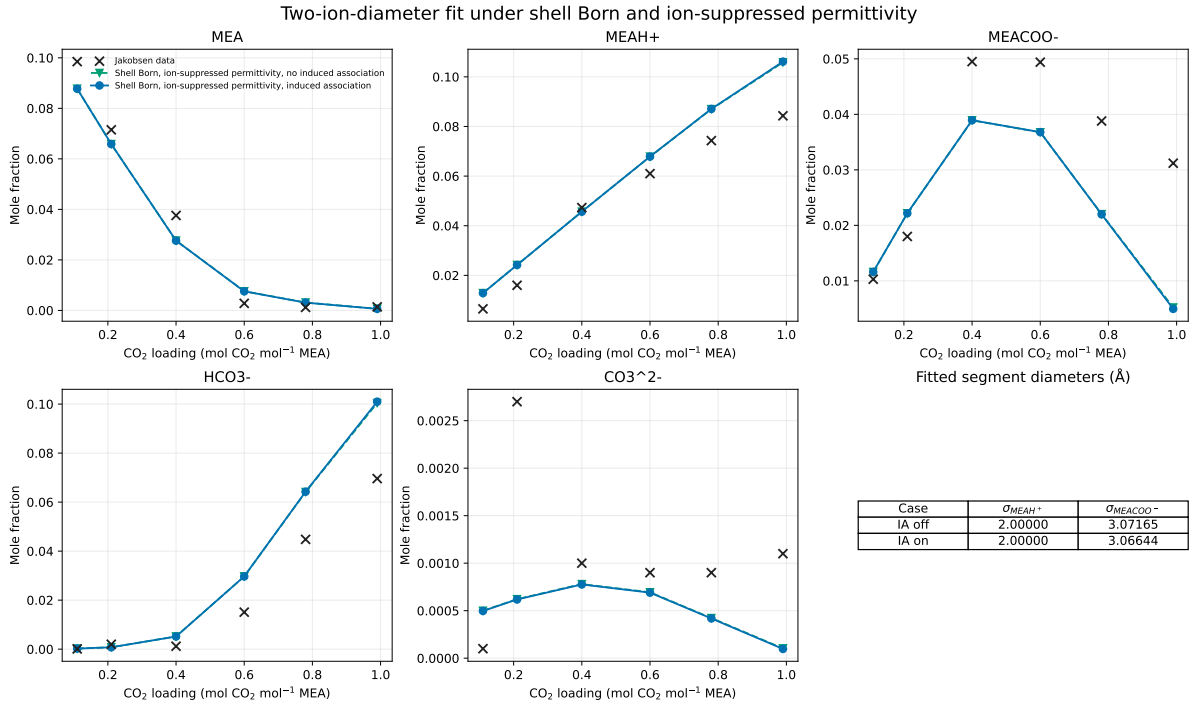


Figure 3: Direct Jakobsen observations and fitted speciation values. The last panel records the exact fitted segment diameters.

Jacobian condition numbers are 3.23 and 3.21, so the boundary direction is not caused by numerical rank deficiency. Induced association remains retained because its separate CO₂–water phase-equilibrium comparison supports it, even though it is nearly neutral in this fixed-pressure speciation study.

These fits do not qualify a final parameter set: they cover one temperature and solvent concentration, use no observation uncertainty weights, and place the protonated-MEA diameter at the tested boundary. They identify the next model/data discrepancy to resolve with broader pressure, speciation, density, or independent ionic evidence.

2.5 Independent ion-suppression coefficient blocks

The coefficient study retains shell Born, ion-suppressed relative permittivity, CO₂–water induced association, $\alpha_{\text{H}_3\text{O}^+} = 9.55$, and $\alpha_{\text{OH}^-} = 13.96$. Unsupported ions start at 7.01. The first block changes α_{MEA^+} to 2.60 and α_{MEACOO^-} to 7.89 independently and together. The second block applies the same independent comparison to bicarbonate and carbonate on the best full-coverage amine base. No coefficient is fitted.

Changing the MEAH⁺ coefficient is the controlling amine-ion perturbation; MEACOO[−] alone is effectively neutral. Starting from MEAH⁺ = 2.60, bicarbonate supplies most of the additional median improvement and carbonate alone is nearly neutral. Changing both carbonate coefficients gives the lowest tested median factor, 1.26912, but increases the worst individual factor from 6.40 for the MEAH⁺-only configuration to 7.85. The median improvement is there-

Table 4: Fixed ion-suppression coefficient sensitivity across all six Jakobsen loading states.

Configuration	Evaluated	Typical factor
Remaining ions 7.01	6/6	1.32380
MEA ⁺ H = 2.60	6/6	1.28178
MEACOO ⁻ = 7.89	6/6	1.32482
MEA ⁺ H = 2.60; MEACOO ⁻ = 7.89	6/6	1.29242
MEA ⁺ H = 2.60; HCO ₃ ⁻ = 7.89	6/6	1.27083
MEA ⁺ H = 2.60; CO ₃ ²⁻ = 7.89	6/6	1.28067
MEA ⁺ H = 2.60; HCO ₃ ⁻ /CO ₃ ²⁻ = 7.89	6/6	1.26912

fore not uniform across species. The MEA⁺H change primarily reduces the molecular-CO₂ and carbonate discrepancies while slightly worsening the already closer MEA⁺H and MEACOO⁻ predictions.

All 42 states evaluate. Maximum balance and reaction-affinity infinity norms are 8.21×10^{-16} and 2.84×10^{-13} . The calculation establishes local sensitivity at 313.15 K and 30 wt% MEA; it does not identify ion-specific coefficients from direct dielectric observations or establish their transfer across temperature and solvent concentration.

2.6 Broader temperature and pressure comparison

The selected ion block was evaluated without regression against 15 source-backed 30 wt% MEA pressure observations spanning 40–120 °C. All structures retain shell Born, ion-suppressed permittivity, and CO₂–water induced association. The third structure replaces constant water permittivity with the existing Archer–Wang temperature approximation.

Table 5: Broad fixed-structure comparison. Pressure errors use natural-log model/observation residuals; Hajj errors are relative-permittivity units.

Configuration	Pressure	ln-RMSE	Factor	Hajj RMSE
Remaining ions 7.01; constant ϵ_w	7/15	6.802	4.398	32.80
Selected ions; constant ϵ_w	9/15	4.948	2.500	25.40
Selected ions; $\epsilon_w(T)$	10/15	0.913	2.226	32.41

The temperature-dependent water input removes the catastrophic pressure-error branches among the evaluated states and gives the best broad pressure metric. Its diagnostic pressure ln-RMSE is 0.913 and its typical factor is 2.226. A representative 40 °C coupled bubble calculation gives 406.37 Pa, compared with 407.49 Pa from the diagnostic liquid-fugacity calculation at the same state. The broad screen remains limited by five non-evaluable states and by extrapolating the reaction correlations above 323.15 K.

The Hajj comparison does not independently identify the ion coefficients. All structures underpredict its absolute fitted static limit, while $\epsilon_w(T)$ provides the expected temperature direction. Direct single-ion permittivity data do not exist for MEA⁺H and MEACOO⁻ in this model convention; analogous-ion support must instead come from mean ionic activity, transfer, and solvation observables.

The best fixed starting document therefore combines $\alpha_{\text{MEA}^+\text{H}} = 2.60$, $\alpha_{\text{MEACOO}^-} = 7.01$, $\alpha_{\text{HCO}_3^-} = \alpha_{\text{CO}_3^{2-}} = 7.89$, $\alpha_{\text{H}_3\text{O}^+} = 9.55$, and $\alpha_{\text{OH}^-} = 13.96$ with the Archer–Wang water $\epsilon(T)$ approximation. Its exact full mapping is retained in the broad result directory as `best.fixed.candidate.parameters.json`; the complete mapping hash is recorded beside it in `summary.json`. This selects the next regression starting structure; it is not yet a final globally fitted parameter set.

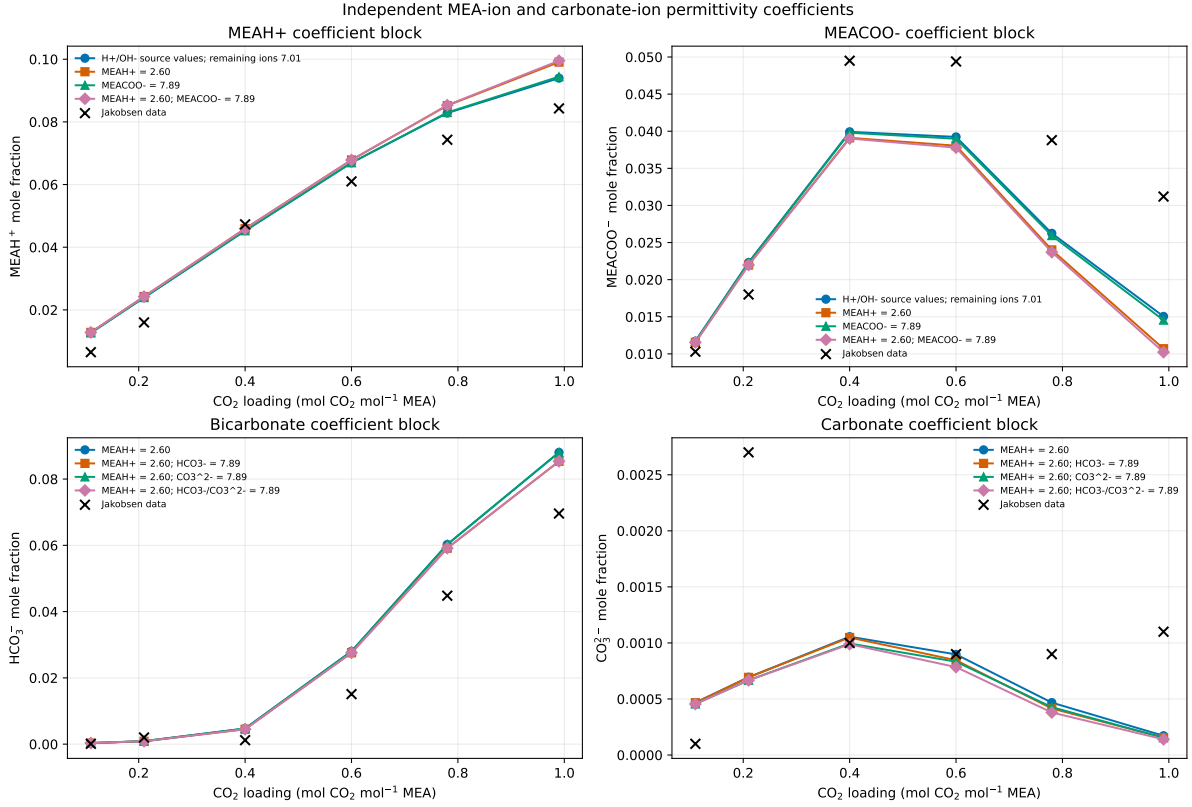


Figure 4: Independent amine-ion and carbonate-ion coefficient changes against the retained Jakobson observations. Exact plotted values are retained in the analysis results.

3 Final shared pressure-speciation refinement

3.1 Question and method

Can one small parameter block improve pressure and direct speciation together after fixing the selected shell-Born, ion-suppressed-permittivity, and CO₂-water induced-association structure? The calculation fits one common $k_{\text{CO}_2, \text{MEA}}$, σ_{MEA^+} , and σ_{MEACOO^-} across all retained states. Each state is a fixed-temperature, fixed-pressure, one-liquid GREPE problem. Molecular-CO₂ liquid fugacity supplies the fast pressure surrogate; direct Jakobson species mole fractions supply the speciation observations. Natural-log residuals are normalized so pressure and speciation have equal block weight.

The installed non-editable Engine wheel has SHA-256:

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e14288867d4fb5bc1367dd0de490aeb
1551f1613074aced0a8d28432ca762f23
```

The primary application domain is 30 wt% MEA at 313.15, 333.15, and 353.15 K. All 59 available pressure observations in those profiles were attempted. Thirty-eight pressure rows and 17 speciation targets from three loading states entered the fit. Twenty-one pressure states and three speciation states failed trace-state certification before the fit and remain explicit failures rather than penalty residuals.

3.2 Result

The complete-row one-step refinement reduced equal-block cost by 42.2%, from 0.9452311 to 0.5460640, in 213 seconds. This is a usable refinement, not a formal optimizer convergence claim. Pressure-surrogate RMS and median factors are 2.114 and 2.046; speciation RMS and

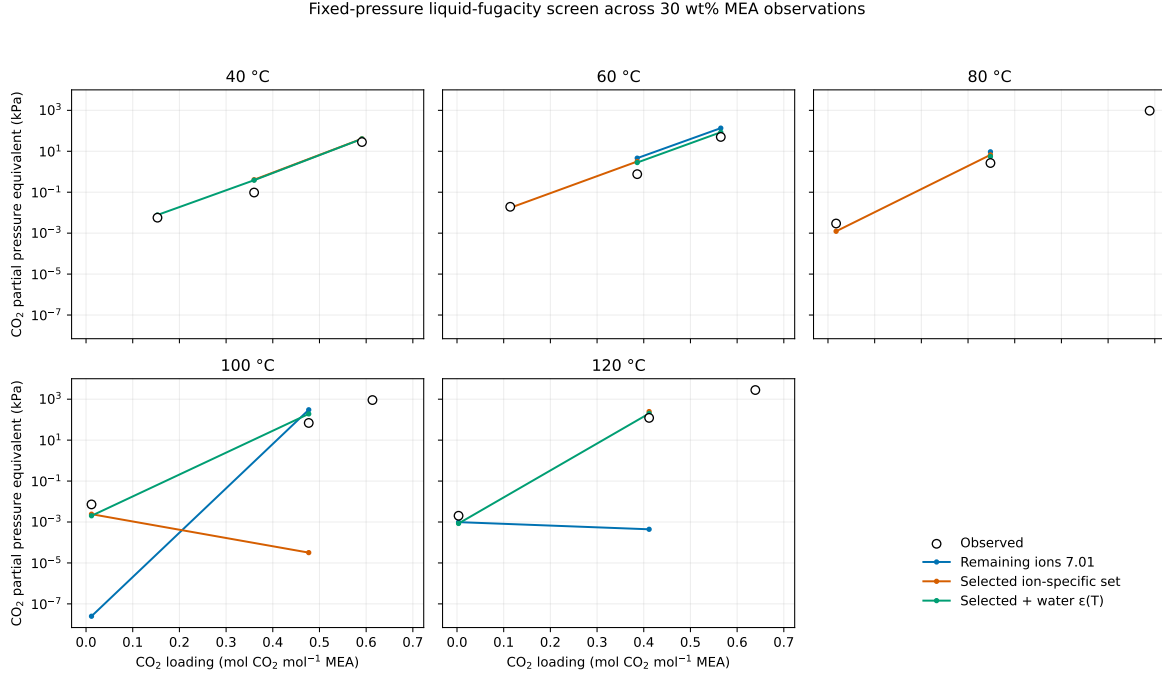


Figure 5: Diagnostic fixed-pressure neutral-CO₂ liquid fugacity across the five temperature profiles. Missing model markers are retained as explicit non-evaluable states.

Table 6: Shared parameter refinement. None of the fitted values contacts its bound.

Parameter	Start	Fitted	Bounds
$k_{\text{CO}_2, \text{MEA}}$	0	-0.03182703	$[-1, 1]$
$\sigma_{\text{MEA}^+\text{H}} \text{ (Å)}$	3.48508557	2.68378739	$[1.5, 5.8]$
$\sigma_{\text{MEA}^-\text{COO}} \text{ (Å)}$	3.53543526	3.26931496	$[1.5, 5.8]$

median factors are 2.073 and 1.540. Scaled-Jacobian singular values are 2.3285, 1.9298, and 0.03822; the condition number is 60.9. No parameter contacts a bound.

3.3 Coupled phase-equilibrium check and interpretation

Three Hilliard states at 40 °C were recomputed with the declared finite liquid and incipient vapor phases. All passed the numerical and physical certification criteria. Predicted CO₂ partial pressures are 3.812, 211.65, and 13928.2 Pa against 5.70, 96.60, and 28300 Pa. The coupled-bubble RMS factor is 1.922.

The mapping is therefore the current usable 30 wt%, 40 °C to 80 °C shared pressure-speciation candidate. The earlier 121-row all-composition and all-temperature attempt is retained separately as a stress-test diagnostic and did not select these parameters. The main remaining limitation is incomplete state coverage. The next Engine work should batch or parallelize independent row evaluations and repair the 24 trace-state failures before adding another adjustable parameter block.

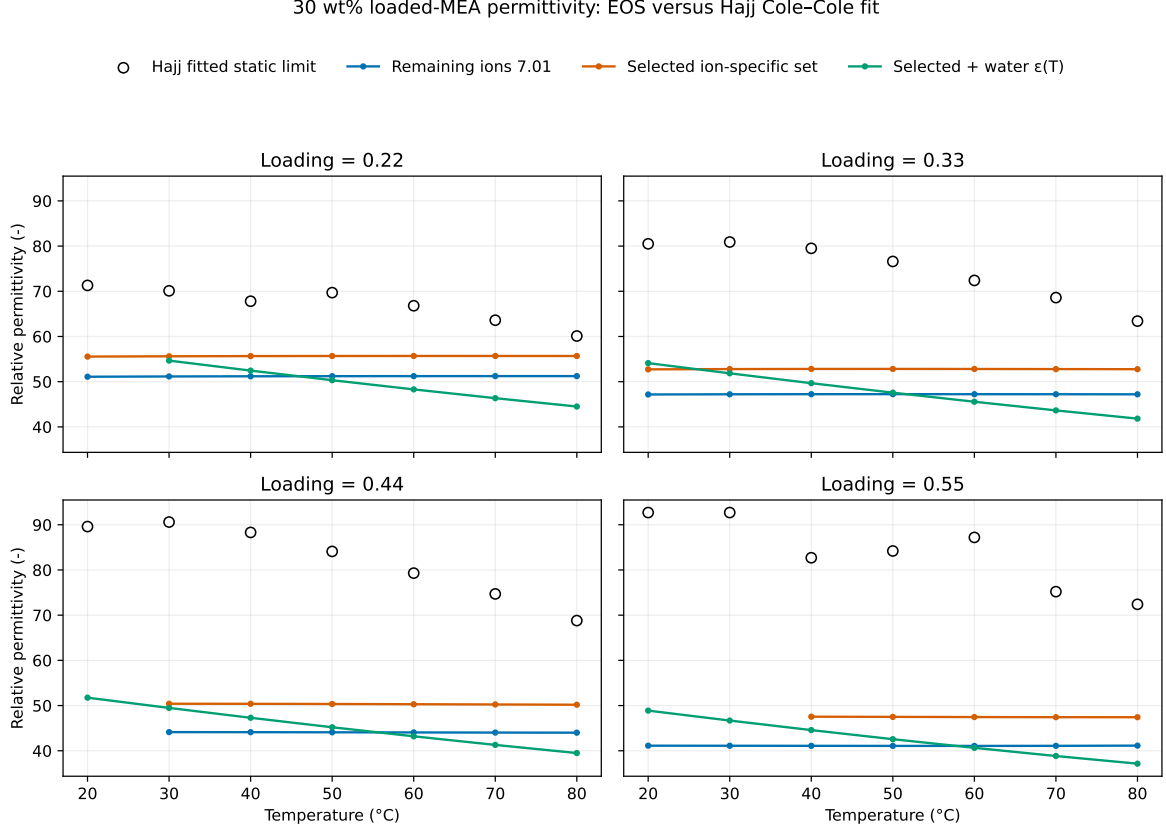


Figure 6: EOS bulk relative permittivity against the Hajj et al. (2024) Cole–Cole fitted static limit. The experimental quantity is an aggregate microwave-fit limit, not a direct DC or single-ion permittivity.

4 Independent pressure–speciation–calorimetry comparison

4.1 Question and method

Does the small loss in former-reserved speciation accuracy outweigh the pressure and calorimetry improvements from the calorimetry-balanced reaction-temperature candidate? The comparison uses frozen evidence outside the parameter fit, but not an untouched independent-validation partition: 27 temperature-challenge pressure rows, 67 former-reserved speciation targets, and 69 direct 30 wt% MEA calorimetry observations. The exact nonlinear candidate replay evaluated all 70 pressure/speciation states without failure.

Errors have different units, and complete observation covariance is not available. Each domain error is therefore normalized by the retained-set error. For nonnegative domain weights w_d summing to one, the comparison score is

$$S = \prod_d \left(\frac{E_{d,\text{balanced}}}{E_{d,\text{retained}}} \right)^{w_d}.$$

A score below one favors the calorimetry-balanced set. The weights represent engineering priorities rather than measurement-uncertainty weights.

4.2 Result

Equal domain weighting gives $S = 0.854$, a 14.6% reduction in combined normalized error. A 0.01-spaced sweep of all 5151 nonnegative weight combinations favors the balanced set for

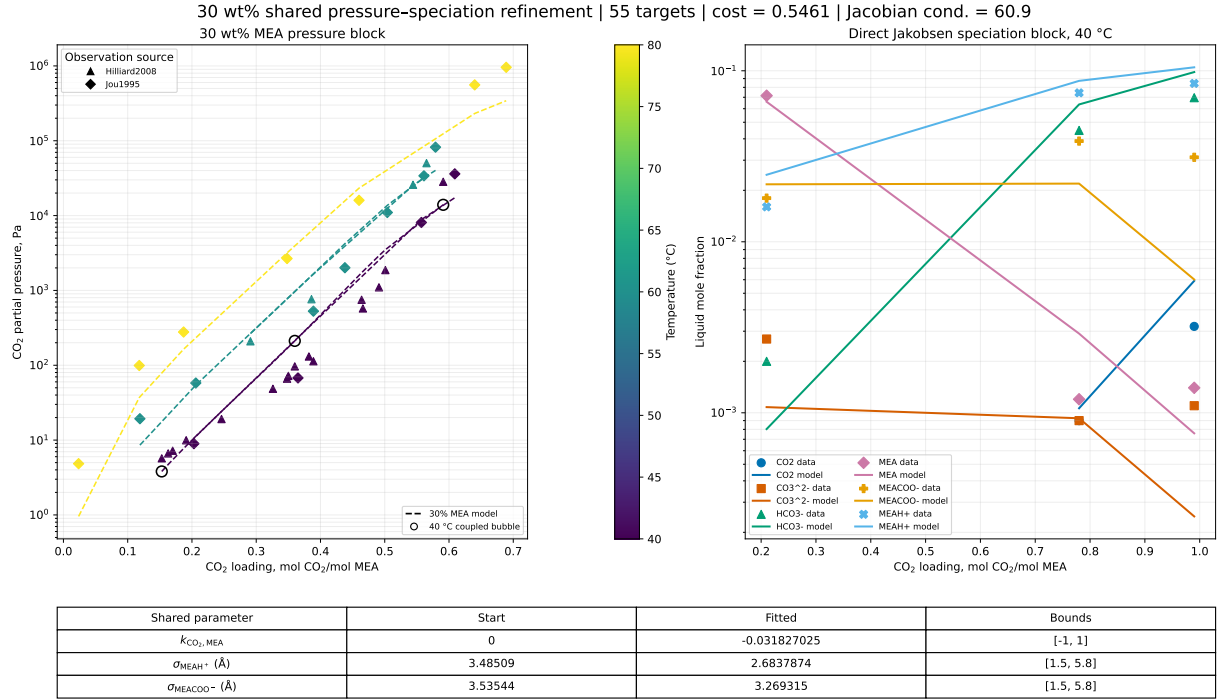


Figure 7: Shared-parameter fit. Points are observations, lines are the fixed-TP pressure surrogate or speciation predictions, and open circles are independent coupled-bubble results at 40 °C. The exact plotted values are retained beside the fit result.

Table 7: Multiproperty errors and balanced-to-retained ratios.

Domain	Retained	Balanced	Ratio
Pressure ln-RMSE	0.7932	0.7133	0.899
Reserved-speciation ln-RMSE	0.8481	0.8830	1.041
Calorimetry RMSE (kJ/mol CO ₂)	15.755	10.489	0.666

97.2% of the grid. The retained set is selected only when reserved speciation receives more than 72.5–91.0% of the total weight; the threshold depends on the pressure/calorimetry split.

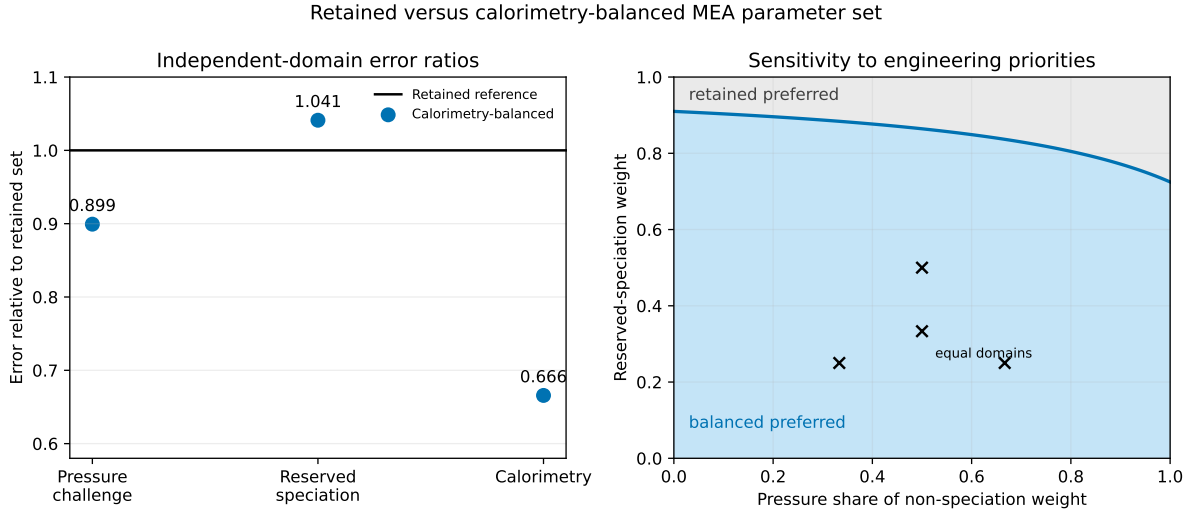


Figure 8: Independent-domain error ratios and sensitivity of the preferred parameter set to engineering priorities. The exact plotted values are retained beside the figure.

4.3 Interpretation and limitations

The calorimetry-balanced candidate has the lower diagnostic score under equal weighting and under each tested single-domain-emphasis profile. The retained set remains preferable only for a decision dominated almost entirely by former-reserved speciation. This ranking does not support parameter adoption or absorber-model transfer.

The comparison does not establish statistical selection. Speciation and calorimetry observations include correlated rows, and full covariance matrices are unavailable. The calorimetry observable is a Gibbs–Helmholtz diagnostic that neglects the pressure–volume correction and represents an average 313.15–353.15 K temperature derivative. Exact inputs and results are retained under `results/independent_evidence_comparison/` in the Born-permittivity analysis directory.

5 Issue 70 parameter decision

Issue 70 ends with a supported negative result. The retained diagnostic candidate evaluated 119 equilibrium states and 257 comparison quantities with no omitted final comparison row, but its parameter optimizer terminated with `NO_CONVERGENCE`. The active scaled Jacobian had condition number 4745.95 and smallest singular value 4.27×10^{-4} . Rows formerly marked reserved were opened during model selection, so no untouched replacement partition remains for independent validation. The candidate calculation also used an Engine wheel whose SHA-256 differs from the frozen Engine lock.

The multiproperty comparisons remain useful diagnostics, not validation. In particular, the Wong molecular-CO2 challenge contains sodium perchlorate beyond the nine-species chemistry, and the calorimetry comparison lacks an accepted reference-state derivative and covariance model. These competing explanations prevent attribution of the residual improvements to one predictive parameter set.

The machine-readable decision record and generated parameter, reaction, comparison, and downstream-property tables are retained in the Issue 70 results directory. The associated refusal record prohibits predictive export and absorber-column mapping. The fixed-parameter manuscript boundary therefore remains in force.